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Lab 2 – TCP/IP (CH4)

Part 1: TCP/IP configuration of your computer

A. Every computer that is on the LAN/Internet and uses TCP/IP needs 4 things. Follow the instructions below and find these 4 things for your computer and explain why and how they are used. **(15 POINTS)**
To enable to view your current network settings:
PC
- Click **Start→Run**, and then type "**cmd**";
- From the Command prompt, type "**ipconfig/all**"

MAC
- Click the **Apple** menu then select **System Preferences**
- Look for **Internet & Wireless** section and click **Network**
- Select **AirPort** and then click **Advanced**
- Click **TCP/IP,** then **Configure IPv4**

B. Take a screen shot and paste below, then use this information to fill out the table below.

#	TCP/IP configuration	Points	Your computer's	Why is this needed? How does your computer obtain this?
1	Its IP address	1	10.106.97.228	This uniquely identifies my computer and is needed to send and receive data. DCHP server provides this.
2	Subnet mask	1	255.255.254.0	This is for routing, inside subnet is decentralized, outside is centralized. DCHP server provides this..
3	IP address of DNS server	1	131.151.245.20 131.151.245.21	Need for IP address resolution. DCHP server provides this.
4	IP address of subnet gateway	1	10.106.97.254	Need to be able to communicate with computers outside of subnet. DCHP server provides this.
5	Your screenshot goes here:	1	Screenshots attached below:	

```
C:\Users\DaneShuler>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DaneMSIGS66
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Broadcast
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : mst.edu

Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : TAP-Windows Adapter V9 for Ope
   Physical Address. . . . . . . . . : 00-FF-7E-76-85-CE
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Killer E3100G 2.5 Gigabit Ethe
   Physical Address. . . . . . . . . : D8-BB-C1-AC-59-21
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet
   Physical Address. . . . . . . . . : 0A-00-27-00-00-10
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::2317:18e0:de38:c770%16(P
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 990511143
```

```
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-29-15-11-60-D8-BB-C1-AC-59
   DNS Servers . . . . . . . . . . . : fec0:0:0:ffff::1%1
                                       fec0:0:0:ffff::2%1
                                       fec0:0:0:ffff::3%1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : A0-E7-0B-CC-5C-9B
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 4:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : A2-E7-0B-CC-5C-9A
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 2:

   Connection-specific DNS Suffix  . : mst.edu
   Description . . . . . . . . . . . : Killer(R) Wi-Fi 6E AX1675x 160MHz Wire
   Physical Address. . . . . . . . . : A0-E7-0B-CC-5C-9A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::abdf:b054:31e7:e781%10(Preferred
   IPv4 Address. . . . . . . . . . . : 10.106.97.228(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.254.0
   Lease Obtained. . . . . . . . . . : Monday, March 3, 2025 3:04:00 PM
   Lease Expires . . . . . . . . . . : Monday, March 3, 2025 5:03:59 PM
   Default Gateway . . . . . . . . . : 10.106.97.254
   DHCP Server . . . . . . . . . . . : 131.151.248.65
   DHCPv6 IAID . . . . . . . . . . . : 111208203
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-29-15-11-60-D8-BB-C1-AC-59
   DNS Servers . . . . . . . . . . . : 131.151.245.20
                                       131.151.245.21
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Networ
   Physical Address. . . . . . . . . : A0-E7-0B-CC-5C-9E
```

Tracert

Traceroute is a computer network tool for measuring the route path and transit times of packets across an Internet Protocol (IP) network.

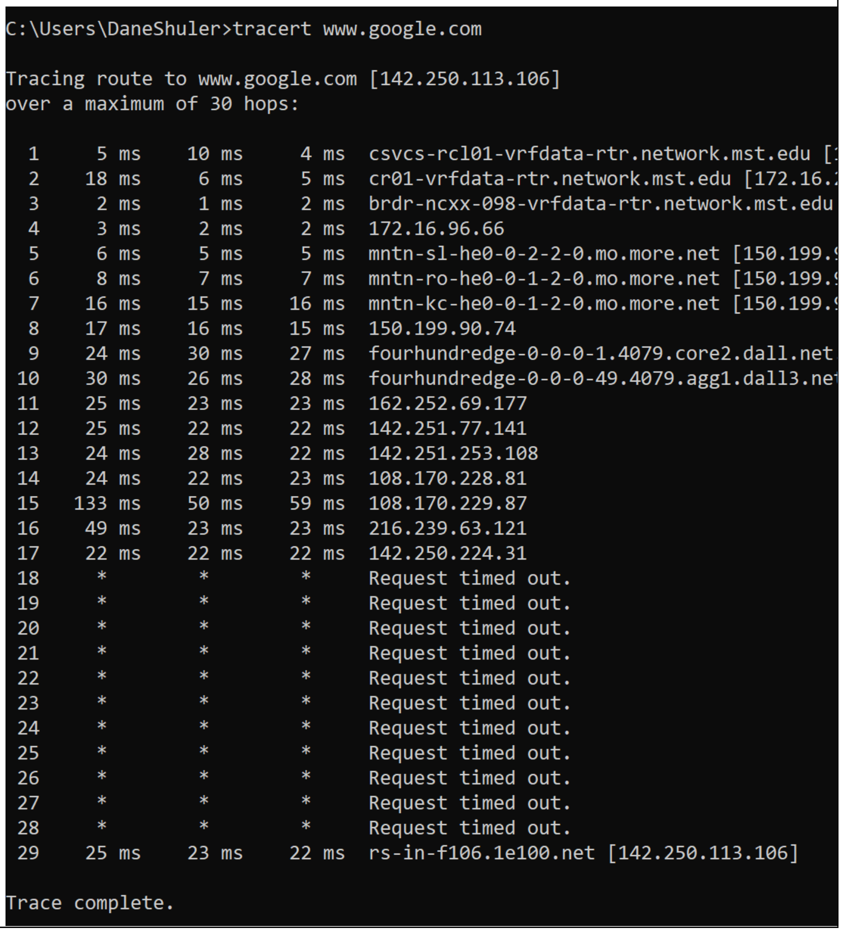
A. To start measuring the route path and transit times of packets across the IP network to the destination url:
   - Click Start→Run, and then type "**cmd**";
   - From the Command prompt, type "**tracert**"

B. Type "**tracert** www.google.com"

C. Wait for the **tracert** command to finish running. Take a screenshot of the results of **tracert** and paste it below.

#	Question	Points	Your answer
6	How many hops (routers) the packet travelled from your computer to the destination (www.google.com)?	1	29 hops
7	What can you tell about the path that the packet travelled?	1	The packet first went to the default gateway, then passed through two routers within the S&T network before exiting to MOREnet and Internet2, ultimately reaching Google's network.
8	Research and explain how can **tracert** command be used for network attacks?	1	Attackers use tracert for network reconnaissance, mapping routers, firewalls, and weak points to plan DDoS, MITM, or routing attacks. ICMP filtering helps mitigate risks.

9	Screenshot of the result of tracert	1	Screenshot attached below:



```
C:\Users\DaneShuler>tracert www.google.com

Tracing route to www.google.com [142.250.113.106]
over a maximum of 30 hops:

  1     5 ms    10 ms     4 ms  csvcs-rcl01-vrfdata-rtr.network.mst.edu [
  2    18 ms     6 ms     5 ms  cr01-vrfdata-rtr.network.mst.edu [172.16.
  3     2 ms     1 ms     2 ms  brdr-ncxx-098-vrfdata-rtr.network.mst.edu
  4     3 ms     2 ms     2 ms  172.16.96.66
  5     6 ms     5 ms     5 ms  mntn-sl-he0-0-2-2-0.mo.more.net [150.199.
  6     8 ms     7 ms     7 ms  mntn-ro-he0-0-1-2-0.mo.more.net [150.199.
  7    16 ms    15 ms    16 ms  mntn-kc-he0-0-1-2-0.mo.more.net [150.199.
  8    17 ms    16 ms    15 ms  150.199.90.74
  9    24 ms    30 ms    27 ms  fourhundredge-0-0-0-1.4079.core2.dall.net
 10    30 ms    26 ms    28 ms  fourhundredge-0-0-0-49.4079.agg1.dall3.ne
 11    25 ms    23 ms    23 ms  162.252.69.177
 12    25 ms    22 ms    22 ms  142.251.77.141
 13    24 ms    28 ms    22 ms  142.251.253.108
 14    24 ms    22 ms    23 ms  108.170.228.81
 15   133 ms    50 ms    59 ms  108.170.229.87
 16    49 ms    23 ms    23 ms  216.239.63.121
 17    22 ms    22 ms    22 ms  142.250.224.31
 18     *        *        *     Request timed out.
 19     *        *        *     Request timed out.
 20     *        *        *     Request timed out.
 21     *        *        *     Request timed out.
 22     *        *        *     Request timed out.
 23     *        *        *     Request timed out.
 24     *        *        *     Request timed out.
 25     *        *        *     Request timed out.
 26     *        *        *     Request timed out.
 27     *        *        *     Request timed out.
 28     *        *        *     Request timed out.
 29    25 ms    23 ms    22 ms  rs-in-f106.1e100.net [142.250.113.106]

Trace complete.
```

Netstat

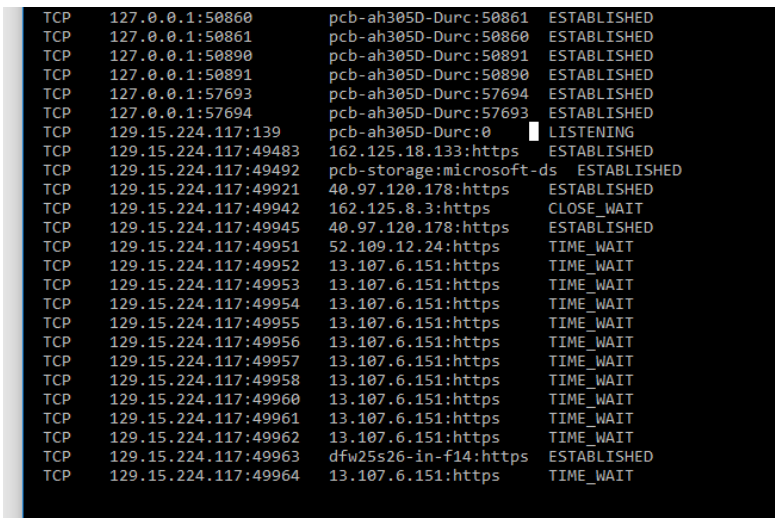
Applications and services that receive and send information from a computer over the network are assigned a number (called a "port) that is used to identify the application or service to the operating system and to network devices such as routers and firewalls. Microsoft Windows operating systems include the "Netstat" utility to provide visibility into the ports in use, which ports (applications and services) are connected remotely, and the application or service that opened the port so a connection could be established over the network.

PC

- Use the command prompt window (cmd) and enter the following command: **netstat -a**

MAC

- Open the Finder
- Select Applications and Utilities
- Open the "Terminal.app"
- Type in the command: "netstat -nr"
- The screenshot will look like this.



#	Question	Points	Your answer
10	Explain 2 entries	1	There is an open connection from 10.106.96.89 to 20.59.87.226 using an HTTPS (encrypted HTTP) connection on port 49504. Another active HTTPS connection exists from 10.106.96.89 to edge-star-shv-02-ord5 on port 60884.
11	How could you use "**netstat**" to detect spyware/malware on your	1	Use netstat -ano to check for suspicious connections, unknown foreign IP addresses, and unusual listening ports. Cross-check process IDs (PIDs) with Task Manager to identify potential malware.

	computer? (Hint: google it)		
12	Screenshot from your computer	1	Screenshot attached below: TCP 10.106.96.89:49504 20.59.87.226:https TCP 10.106.96.89:60857 170.114.15.222:htt TCP 10.106.96.89:60858 20.59.87.226:https TCP 10.106.96.89:60874 134.224.240.218:ht TCP 10.106.96.89:60877 162.159.136.234:ht TCP 10.106.96.89:60878 edge-star-shv-02-c TCP 10.106.96.89:60882 edge-dgw-shv-02-or TCP 10.106.96.89:60884 edge-star-shv-02-c TCP 10.106.96.89:60901 edge-star-shv-02-c

Routing Table

Routing tables are an important part of TCP/IP protocol stack.

PC

- Use the command prompt window (cmd; Terminal.app on a Mac) and enter the following command: **route print**.

MAC

- Type in the command: “**netstat -nr**”

Our goal is to look at the IPv4 Route Table such as the one below:

```
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0     129.15.224.1   129.15.224.117     25
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
     129.15.224.0    255.255.252.0         On-link    129.15.224.117    281
   129.15.224.117  255.255.255.255         On-link    129.15.224.117    281
   129.15.227.255  255.255.255.255         On-link    129.15.224.117    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link    129.15.224.117    281
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link    129.15.224.117    281
===========================================================================
Persistent Routes:
  None
```

#	Question	Points	Answer
13	Your laptop’s routing table	1	Routing table below

```
Interface List
 15...00 ff 7e 76 85 ce ......TAP-Windows Adapter V9 for OpenVPN Connect
 19...d8 bb c1 ac 59 21 ......Killer E3100G 2.5 Gigabit Ethernet Controller
 16...0a 00 27 00 00 10 ......VirtualBox Host-Only Ethernet Adapter
 25...a0 e7 0b cc 5c 9b ......Microsoft Wi-Fi Direct Virtual Adapter #3
 14...a2 e7 0b cc 5c 9a ......Microsoft Wi-Fi Direct Virtual Adapter #4
 10...a0 e7 0b cc 5c 9a ......Killer(R) Wi-Fi 6E AX1675x 160MHz Wireless Network
  3...a0 e7 0b cc 5c 9e ......Bluetooth Device (Personal Area Network) #2
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    10.106.97.254    10.106.96.89     50
      10.106.96.0    255.255.254.0         On-link     10.106.96.89    306
     10.106.96.89  255.255.255.255         On-link     10.106.96.89    306
    10.106.97.255  255.255.255.255         On-link     10.106.96.89    306
        127.0.0.0        255.0.0.0         On-link        127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link        127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link        127.0.0.1    331
     192.168.56.0    255.255.255.0         On-link     192.168.56.1    281
     192.168.56.1  255.255.255.255         On-link     192.168.56.1    281
   192.168.56.255  255.255.255.255         On-link     192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link        127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link     192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link     10.106.96.89    306
  255.255.255.255  255.255.255.255         On-link        127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link     192.168.56.1    281
  255.255.255.255  255.255.255.255         On-link     10.106.96.89    306
===========================================================================
Persistent Routes:
  None
```

14	The very first entry in the routing table is the Default Route. This route has a network destination of 0.0.0.0 and Netmask of 0.0.0.0. What is the	1	10.106.97.254 This is the default route because it directs all traffic that does not have a more specific route in the table. The system uses this route to reach external networks, including the internet, via the gateway. Routers handle multiple routes to connect different networks efficiently.

	Gateway and Interface for the default route? Why?		
15	What is the purpose of Metric column in the routing table?	1	The Metric column in the routing table indicates the cost or priority of a route. Lower values represent preferred routes, meaning the system will use the route with the smallest metric when multiple paths to the same destination exist.